

DSA-210 Final Report

Dance Genre Classification from Motion-Based Features Using the AIST++ Dataset

1. Motivation

In this project, I studied whether dance movements can be used to classify dance genres. Dance is naturally a structured form of movement, and different genres often have their own motion style, intensity, and rhythm. Because of this, I wanted to see whether these differences could be captured numerically and then used in a machine learning classification task.

I was interested in this topic for two reasons. First, it is a meaningful data science problem because it combines real-world sequential data, feature extraction, statistical testing, and machine learning. Second, dance genre classification is not a trivial task. Even if two genres are visually different to humans, their motion patterns may overlap when they are reduced to a small set of numeric features. This makes the project a good example of how data representation affects machine learning performance.

In the proposal stage, I originally planned to explore both motion-based and music-based information. The goal was not only to test whether dance genres could be identified from motion data, but also to examine whether adding music-related features such as tempo and rhythm might improve the results. In the final implementation, however, I focused on the motion side of the problem and developed a complete pipeline around motion-derived features.

2. Data Source

I used the publicly available AIST++ dance dataset as the main data source. In the proposal, I selected this dataset because it contains dance motion data from multiple genres and is therefore suitable for genre-based comparison and classification.

For the implemented version of the project, I worked mainly with the motion files and extracted features from `simpl_trans`, which represents global body translation over time. According to the repository progress notes, the project used AIST++ motion data, extracted simple motion-based features from `simpl_trans`, and then applied exploratory analysis, hypothesis testing, and machine learning models.

The raw motion files and ignore list were used locally during preprocessing and cleaning and were not uploaded to the repository. The executed notebook outputs were kept in the repository to document the milestone workflow and results.

3. Data Preparation and Feature Extraction

The first stage of the project consisted of collecting the motion files, reading the relevant sequence data, and constructing a structured feature table for analysis. The final processed dataset contained 411 motion sequences and 8 columns, with no missing values in the derived feature table. This made it suitable for both statistical analysis and supervised machine learning.

The target label was the dance genre. The features extracted for each sequence were:

- `num_frames`
- `mean_motion`
- `std_motion`
- `max_motion`
- `median_motion`
- `total_displacement`

These features were derived from the frame-to-frame differences in `simpl_trans`. In other words, I transformed the raw movement trajectory into summary measures that represent how much the dancer moved, how variable that movement was, how large the maximum movement step was, and how much total movement accumulated over the sequence.

This feature design intentionally simplified the original motion signal. Instead of feeding the raw sequence directly into a model, I created interpretable summary variables that could be analyzed statistically first and then used for classification. This approach matched the overall workflow of the project, which included data collection, preprocessing, EDA, statistical analysis, machine learning, and reporting.

4. Exploratory Data Analysis

After creating the processed feature table, I conducted exploratory data analysis to understand the structure of the dataset and the general patterns in the motion features.

The EDA stage focused on three main questions. First, I examined the genre distribution to understand whether the dataset was reasonably balanced across classes. Second, I looked at sequence lengths through `num_frames` to see whether some genres tended to have longer or shorter sequences. Third, I explored motion-based variables such as average motion and motion variability to see whether there were visible differences between genres.

The exploratory analysis suggested that the dataset contained multiple dance genres with different movement characteristics. The repository summary also notes that the project included exploratory analysis on genre distribution and sequence lengths as well as visualization of motion patterns across genres.

From the plots and grouped summaries, some genres appeared more intense in movement while others were relatively more stable or contained lower average motion. However, the EDA also suggested that there was overlap. This was important because it already hinted that classification might be possible, but not easy.

5. Hypothesis Testing and Statistical Analysis

To move beyond visual inspection, I applied statistical tests to check whether the extracted motion features differed significantly across genres.

The main hypothesis testing tool was ANOVA. I first tested whether `mean_motion` differed significantly across dance genres. The result was highly significant, indicating that not all genres had the same average movement intensity. This was one of the strongest findings in the EDA milestone and supported the idea that motion-derived variables do contain genre-related information.

I then extended the statistical analysis by testing additional motion features. ANOVA was also applied to:

- `std_motion`
- `max_motion`
- `median_motion`
- `total_displacement`

These additional tests were also significant. This suggests that genre differences are not limited to average movement alone. Dance genres also differ in movement variability, peak movement intensity, typical step magnitude, and total movement over time.

I also examined relationships between features using correlation analysis. Two patterns were especially important.

First, `num_frames` and `total_displacement` were strongly positively correlated. This means that longer sequences naturally tend to accumulate more total movement, so `total_displacement` should not be interpreted as a purely style-based variable. Part of it may reflect sequence length rather than only dance genre.

Second, `mean_motion` and `std_motion` were also strongly positively correlated. This suggests that sequences with higher average movement intensity also tend to show greater variability in movement. In other words, these features are informative, but they are not fully independent.

This statistical stage was important for two reasons. It gave evidence that motion features differ across classes, and it also helped explain why some features might be more useful than others in the later machine learning step.

6. Machine Learning Methods

After EDA and hypothesis testing, I applied supervised machine learning methods to predict dance genres from the extracted motion-based features.

The target variable `genre` was encoded as numeric labels. The feature set consisted of the six motion-derived variables listed earlier. The data was split into training and test sets using an 80/20 split with stratification so that class proportions were preserved across the split.

I trained and compared three classification models:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Random Forest

This model comparison is also described in the repository summary, which states that Logistic Regression, KNN, and Random Forest were trained and compared, and that the models were evaluated using accuracy and a confusion matrix.

The purpose of using multiple models was to compare a linear baseline with nonlinear alternatives. Logistic Regression served as a simple baseline. KNN provided a distance-based nonlinear method. Random Forest provided a more flexible ensemble-based model that could capture nonlinear interactions between features.

7. Results and Findings

The machine learning results showed that the problem is partially learnable from motion-derived features, but not solved well by a small summary-feature representation.

The approximate test accuracies were:

- Logistic Regression: about 0.35
- KNN: about 0.40
- Random Forest: about 0.48

Random Forest achieved the best performance among the three models, followed by KNN and then Logistic Regression. This suggests that nonlinear models are more suitable for this task than a simple linear baseline.

The confusion matrix for the Random Forest model showed that some genres were predicted more accurately than others, while several genres were still confused with one another. This is an important result. It indicates that the extracted motion features are meaningful and do contain genre-related signal, but the information is not strong enough to separate all classes clearly.

The repository notes already summarize this general conclusion by stating that initial results suggest average motion differs significantly across genres and that motion-based features may be useful for dance genre classification. My final results support that statement, but with an important qualification: the features are useful, yet still limited.

8. Discussion

The most important discussion point in this project is not that the accuracy was high, but that the full pipeline revealed both the strengths and limitations of the current feature design.

The EDA and hypothesis tests showed that genres do differ significantly in several motion-based variables. This means the dataset is not random with respect to genre labels. There is real structure in the movement data.

However, the machine learning results showed that this structure is only partially captured by the features I used. The models were able to perform better than a weak baseline, and Random Forest clearly outperformed Logistic Regression, but the overall accuracy remained modest. This suggests that the classification task is more complex than what can be represented by a few global summary statistics.

In my interpretation, the low-to-moderate performance has several likely reasons:

First, the feature set is simple. It compresses each dance sequence into only a few aggregate values. This removes detailed temporal structure, ordering information, and joint-level motion patterns.

Second, some genres may overlap in terms of movement intensity and global displacement. Two genres can differ stylistically but still have similar average movement or total motion.

Third, some features are correlated with each other. For example, `total_displacement` is strongly related to sequence length, and `mean_motion` is strongly related to `std_motion`. This reduces the diversity of information available to the model.

Fourth, the original proposal included the idea of combining motion information with music-related features, but the final implementation focused on motion features only. It is possible that multimodal enrichment could have improved classification by adding complementary information.

9. Limitations and Future Work

This project has several limitations.

The most important limitation is the restricted feature representation. I used only motion-derived summary features from `simpl_trans`, which means the model did not directly use detailed pose dynamics, local body movement, or richer temporal patterns.

A second limitation is that I did not complete the original enrichment idea from the proposal. In the proposal, I planned to combine motion data with additional music-based features such as tempo and rhythm in order to compare motion-only and motion-plus-music models. In the final implemented workflow, I focused on motion features only. This means the final project became a scoped-down version of the original plan.

A third limitation is that classification accuracy remained below 0.50 even for the best-performing model. This does not make the project unsuccessful, but it does show that the current feature set is not sufficient for strong genre prediction.

In future work, several improvements could be explored:

- extracting richer pose-based features rather than only global translation
- using temporal descriptors that preserve motion sequence structure
- incorporating music-based variables such as tempo, beat, or rhythm
- performing more systematic hyperparameter tuning
- trying additional models that can better handle sequence-like data
- analyzing class confusion in more depth to understand which genres are intrinsically similar

If I continued this project, the most important next step would be to enrich the representation rather than just switching classifiers. The current results suggest that the main bottleneck is not only model choice, but also feature choice.

10. Conclusion

This project investigated whether dance genres can be classified from movement data using the AIST++ dataset. The final workflow included data preprocessing, motion-based feature extraction, exploratory data analysis, statistical hypothesis testing, and supervised machine learning classification.

The findings showed that motion-based features differ significantly across genres and therefore carry meaningful information. However, the machine learning results also showed that a small set of summary features is not enough for strong classification performance. Among the tested models, Random Forest performed best, but overall accuracy remained moderate.

My final conclusion is that dance genre classification from motion data is possible to some extent, but the success of the task depends heavily on feature design. Motion-only summary variables provide a useful starting point, yet they do not fully represent the complexity of dance style. This makes the project valuable not because it achieved very high accuracy, but because it clearly shows the connection between data representation, statistical structure, and model performance.

11. AI Usage Disclosure

I used AI tools during this project for support in brainstorming structure, improving wording, debugging some implementation issues, and organizing parts of the written report. However, the dataset selection, preprocessing, feature extraction, notebook execution, hypothesis testing, model training, result inspection, and final interpretation of the outputs were carried out by me. I reviewed and edited the final text and kept the final analytical decisions under my own control, in line with the course requirement that AI use must be explicitly documented.